

Rules of Inference
Arguments
Introduction to Proofs

Lecture-3

Logic and Proofs



Arguments in Propositional Logic

- A *argument* in propositional logic is a sequence of propositions. All but the final proposition are called *premises*. The last statement is the *conclusion*.
- The argument is valid if the premises imply the conclusion. An *argument form* is an argument that is valid no matter what propositions are substituted into its propositional variables.
- If the premises are $p_1, p_2, \dots, p_n$ and the conclusion is q then
 $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology.
- This truth table show $\neg p \vee q$ is equivalent to $p \rightarrow q$.

- Inference rules are all argument simple argument forms that will be used to construct more complex argument forms.

- Inference rules are all argument simple argument forms that will be used to construct more complex argument forms.
- Rules of Inference for Propositional Logic:
(1) Modus Ponens

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

Corresponding Tautology:

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

Example:

if you read well then you will get a job.

you are reading well then the conclusion is you will get the job.

(2) Modus Tollens

$$\begin{array}{c} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

Corresponding Tautology:

$$(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$$

- **Example:**
- Let p be “it is snowing.”, Let q be “I will study discrete math.”
- “If it is snowing, then I will study discrete math.”
- “I will not study discrete math.”
- “Therefore , it is not snowing.”

(3) Hypothetical Syllogism

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Corresponding Tautology:
 $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

- **Example:**
- Let p be “it snows.”, Let q be “I will study discrete math.”
- Let r be “I will get an A.”
- “If it snows, then I will study discrete math.”
- “If I study discrete math, I will get an A.”
- “Therefore , If it snows, I will get an A.”

(4) Disjunctive Syllogism

$$\frac{p \vee q \quad \neg p}{\therefore q}$$

Corresponding Tautology:
 $(\neg p \wedge (p \vee q)) \rightarrow q$

- **Example:**
- Let p be “I will study discrete math.”
Let q be “I will study English literature.”
- “I will study discrete math or I will study English literature.”
- “I will not study discrete math.”
- “Therefore , I will study English literature.”

(5) Addition

$$\frac{p}{\therefore p \vee q}$$

Corresponding Tautology:

$$p \rightarrow (p \vee q)$$

- **Example:**
- Let p be “I will study discrete math.”
Let q be “I will visit Las Vegas.”
- “I will study discrete math.”
- “Therefore, I will study discrete math or I will visit Las Vegas.”

(6) Simplification

$$\frac{p \wedge q}{\therefore q}$$

Corresponding Tautology:

$$(p \wedge q) \rightarrow p$$

- **Example:**
- Let p be “I will study discrete math.”
Let q be “I will study English literature.”
- “I will study discrete math and English literature”
- “Therefore, I will study discrete math.”

(7) Conjunction

$$\frac{p \quad q}{\therefore p \wedge q}$$

Corresponding Tautology:
 $((p) \wedge (q)) \rightarrow (p \wedge q)$

- **Example:**
- Let p be “I will study discrete math.”
Let q be “I will study English literature.”
- “Therefore, I will study discrete math and I will study English literature.”

(8) Resolution

$$\frac{\neg p \vee r \quad p \vee q}{\therefore q \vee r}$$

Corresponding Tautology:

$$((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$$

- **Example:**
- Let p be “I will study discrete math.”
Let r be “I will study English literature.”
Let q be “I will study databases.”
- “I will not study discrete math or I will study English literature.”
- “I will study discrete math or I will study databases.”
- “Therefore, I will study databases or I will English literature.”

Valid Arguments

- A *valid argument* is a sequence of statements. Each statement is either a premise or follows from previous statements by rules of inference. The last statement is called conclusion.
- A valid argument takes the following

S_1
S_2
S_3
S_4
.
.
$\therefore C$

- **Example 1:** From the single proposition

$$p \wedge (p \rightarrow q)$$

Show that q is a conclusion.

- **Solution:**

$$p \wedge (p \rightarrow q)$$

premise

$$p$$

Conjunction

$$(p \rightarrow q)$$

Conjunction

$$(p \rightarrow q)$$

Modus Ponens

Test the validity of the following argument

Q.1

$$\begin{array}{l} p \rightarrow q \\ \underline{r \rightarrow \neg q} \\ p \rightarrow \neg r \end{array}$$

Q.2

$$\begin{array}{l} p \rightarrow r \\ \neg p \rightarrow q \\ \underline{q \rightarrow s} \\ \neg r \rightarrow s \end{array}$$

Test the validity of the following argument

Q.1

$$(p \vee q) \rightarrow s$$

$$s \rightarrow r$$

$$\underline{\neg(r \vee q)}$$

$$\neg p$$

Example-2

- With these hypotheses:
 - “It is not sunny this afternoon and it is colder than yesterday.”
 - “We will go swimming only if it is sunny.”
 - “If we do not go swimming, then we will take a canoe trip.”
 - “If we take a canoe trip, then we will be home by sunset.”
- Using the inference rules, construct a valid argument for the conclusion:
 - “We will be home by sunset.”

Solution:

1. Choose propositional variables:

p : "It is sunny this afternoon."

r : "We will go swimming."

t : "We will be home by sunset."

q : "It is colder than yesterday."

s : "We will take a canoe trip."

Solution

Counti.....

Translation into propositional logic:

$$\neg p \wedge q$$

$$r \rightarrow p$$

$$\neg r \rightarrow s$$

$$s \rightarrow t$$

And conclusion :t

Solution

Counti.....

Translation into propositional logic:

$$\neg p \wedge q$$

Premise

$$\neg p$$

Simplification

$$r \rightarrow p$$

Premise

$$\neg r$$

Modus tollens

$$\neg r \rightarrow s$$

Premise

$$s$$

Modus ponens

$$s \rightarrow t$$

Premise

$$t$$

Modus ponens

Valid Arguments

Counti.....

Handling Quantified Statements

Universal Instantiation

$$\frac{\forall(x)P(x)}{\therefore P(c)}$$

Example:

Our domain consists of all dogs and Fido is a dog.

“All dogs are cuddly.”

“Therefore, Fido is cuddly.”

Valid Arguments

Counti.....

Handling Quantified Statements

Universal Generalization (UG)

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$$

Valid Arguments

Counti.....

Handling Quantified Statements

Existential Instantiation (EI)

$$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$$

Valid Arguments

Counti.....

Handling Quantified Statements

Existential Generalization (EG)

$$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$$

Introduction to Proofs

- A proof is a valid argument that establishes the truth of a statement.

Proofs have many practical applications:

- verification that computer programs are correct
- establishing that operating systems are secure
- showing that system specifications are consistent

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Direct Proofs

Example: Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Direct Proofs

Example: Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

Proof: Assume that n is odd. Then $n = 2k + 1$ for an integer k .

$$\begin{aligned}\Rightarrow n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2r + 1\end{aligned}$$

Where $r = 2k^2 + 2k$, an integer

We have proved that if n is an odd integer, then n^2 is an odd integer.

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Direct Proof

Prove that the sum of two rational numbers is rational.

Definition: The real number r is *rational* if there exist integers p and q where $q \neq 0$ such that $r = p/q$

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Direct Proof

Prove that the sum of two rational numbers is rational.

Definition: The real number r is *rational* if there exist integers p and q where $q \neq 0$ such that $r = p/q$

Proof: Assume r and s are two rational numbers. Then there must be integers p, q and also t, u such that

$$r = \frac{p}{q}, s = \frac{t}{u}, u \neq 0, q \neq 0$$

$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu + qt}{qu} = \frac{v}{w}$$

Where $v = pu + qt$, and $w = qu$

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Indirect Proof

- Proof of the contrapositive

- Assume $\neg q$ and show $\neg p$ is true also. This is sometimes called an *indirect proof* method. If we give a direct proof of $\neg q \rightarrow \neg p$ then we have a proof of $p \rightarrow q$.
Why does this work?

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Indirect Proof

- Proof of the contrapositive

- Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Indirect Proof

- Proof of the contrapositive

Solution: Assume n is even. So, $n = 2k$ for some integer k . Thus

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1) = 2j$$

for $j = 3k + 1$

Therefore $3n + 2$ is even. Since we have shown $\neg q \rightarrow \neg p$, $p \rightarrow q$ must hold as well. If n is an integer and $3n + 2$ is odd (not even), then n is odd (not even).

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Indirect Proof

- Proof of the contrapositive

Prove that for an integer n , if n^2 is odd, then n is odd.

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Indirect Proof

- Proof of the contrapositive

Solution: Use proof by contraposition. Assume n is even (i.e., not odd). Therefore, there exists an integer k such that $n = 2k$. Hence,

$$n^2 = 4k^2 = 2(2k^2)$$

and n^2 is even (i.e., not odd).

We have shown that if n is an even integer, then n^2 is even. Therefore by contraposition, for an integer n , if n^2 is odd, then n is odd.

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

- Indirect Proof
 - Proof by Contradiction

Use a proof by contradiction to give a proof that $\sqrt{2}$ is irrational.

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

- Indirect Proof
 - Proof by Contradiction

Solution: Suppose $\sqrt{2}$ is rational. Then there exists integers a and b with $\sqrt{2} = a/b$, where $b \neq 0$ and a and b have no common factors. Then

Therefore a^2 must be even. If a^2 is even then a must be even (an exercise). Since a is even, $a = 2c$ for some integer c . Thus,

Therefore b^2 is even. Again then b must be even as well.

But then 2 must divide both a and b . This contradicts our assumption that a and b have no common factors. We have proved by contradiction that our initial assumption must be false and therefore $\sqrt{2}$ is irrational.